Differential High Order Control Barrier Function-Based Safe Reinforcement Learning

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***Abstract*—Safe reinforcement learning (RL) aims to learn policy while also ensuring the safety constraints. An increasingly common approach is to design a safety filter based on control barrier function (CBF) or high order control barrier function (HOCBF) for the RL policy. A quadratic programming (QP) is then formulated and solved to modify the RL policy, enabling safe exploration. However, directly integrating the safety filter with RL presents two challenges: (1) the conservativeness of safe policy, and (2) potential infeasibility of the QP under bounded input constraints. These issues limit the performance of safe RL. In this letter, we introduce a differential HOCBF constraint by incorporating neural network-based penalty functions into HOCBF. Furthermore, we propose a differential HOCBF-based safe RL framework in which the penalty functions and RL policy are trained concurrently. To address conservativeness, we train penalty functions to maximize long-term rewards while preventing abrupt changes in safe action, thereby achieving ideal performance. To ensure the feasibility of the formulated QP under bounded input constraints, we calculate a set for penalty functions and prove that the feasibility is guaranteed if the learned penalty functions remain within the set. Finally, we verify the effectiveness of the proposed framework on the wheeled mobile robot navigation and obstacle avoidance task.**

***Index Terms*—Reinforcement learning (RL), robot safety, collision avoidance.**

I. Introduction

REINFORCEMENT learning (RL) has been extensively studied for learning control policies in robotic systems and has demonstrated promising performance [1], [2], [3]. An RL agent continuously explores an environment to learn an optimal policy that maximizes a long-term reward. However, for the system subject to safety constraints, unrestricted exploration may cause the agent to access unsafe regions during the training process. Unsafe behaviors could damage the system or pose risks to humans, particularly when deploying RL algorithms on hardware.

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Safe RL aims to learn an optimal policy that maximizes long-term rewards while ensuring hard safety constraints during the learning and deployment processes [4]. Previous literatures have proposed many approaches to learn a safe policy [5], [6], [7], but these approaches do not guarantee the safety during training. Control barrier function (CBF) is an effective way to ensure safety in safety-critical control [8]. To handle the safety constraint of high relative degree, high order control barrier function (HOCBF) is proposed in [9]. By combining the CBF with RL, a CBF-based safe RL framework is proposed in [10]. In this framework, a safety filter based on CBF is designed for RL policy. A quadratic programming (QP) is then formulated and solved to modify the RL policy, achieving safe exploration. This framework can be easily extended to HOCBF-based safe RL to handle safety constraints of high relative degree. The advantages of this framework include: first, safety filters can often be integrated with existing RL algorithms with minimal modifications; second, safety during training is guaranteed by CBF theory. Along this line, A sample-efficient safe RL framework is proposed in [11] to guide safe exploration. Safe RL using disturbance-observer-based CBF is designed for autonomous driving navigation control [12]. A new type of robust CBF-based safe RL is applied in autonomous vehicles trajectory tracking under uncertain [13].

However, directly integrating the CBF (HOCBF) with RL presents two main drawbacks. First, as highlighted in [14], [15], the manually designed class $\mathcal{K}$ function in CBFs may lead to conservativeness problem, which limits the performance of safe RL. Specifically, an over-conservative CBF constraint guides the agent to learn a safe but conservative policy, resulting in low rewards. On the other hand, a least-conservative CBF constraint leads the agent to output an abrupt action signal when it approaches the safety constraint bound. The abrupt signal is hard for actuators to track accurately, increasing the risk of safety constraint violations and potential damage to actuators. This conservativeness issue is even more pronounced in HOCBF constraint [14], [15]. Second, the formulated QP may become infeasible when both the safety constraint and bounded input constraints, such as limits on acceleration and angle velocity, are simultaneously considered [16].

To address the conservativeness, [15] introduces a differentiable HOCBF layer, which is trained via supervised learning on a dataset generated by a nominal optimal controller (e.g., MPC), enabling the HOCBF-based controller to achieve near-optimal control performance. However, designing a nominal optimal controller for a general nonlinear system is challenging, and constructing an optimal control dataset is costly. To ensure the feasibility of the QP, previous works [10], [11], [15] introduce a slack variable stem in the QP, but this may lead to potential

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safety constraint violations. In the HOCBF-based safe RL field, few studies consider the feasibility of the formulated QP under both the safety constraint and bounded inputs constraint.

Based on the above analysis, we propose the differential HOCBF-based safe RL framework to address the conservativeness and feasibility. In the framework, penalty functions represented by a neural network (NN) are introduced into HOCBF constraint to construct differential HOCBF-based QP. By concurrently training penalty functions and safe RL policy, the approach achieves ideal performance while ensuring the feasibility of the formulated QP. The main contributions of this letter are as follows:

- We address the conservativeness by training the penalty functions in differential HOCBF to maximize long-term rewards while preventing abrupt changes in safe action, thereby achieving ideal safe RL performance. Compared with previous work addressing the conservativeness [14], [15], complex auxiliary dynamics and nominal optimal controller which is unavailable in RL scenarios are not required in the proposed method. Penalty functions are adaptively learned using data naturally generated during safe RL exploration, making the method readily deployable.
- In the previous CBF-based safe RL work [10], [11], the feasibility of the formulated QP under both the safety constraint and bounded input constraint is not considered. In this letter, we calculate a set for penalty functions, and demonstrate that the feasibility is guaranteed if the learned penalty functions remain within this set. To our knowledge, this is the first work that systematically considers this issue and provides rigorous feasibility proof. Furthermore, we prove that the policy learned through the proposed framework ensures safety during both training and deployment processes.
- The proposed differential HOCBF-based QP is an expandable framework that can be readily integrated with other policy gradient-based RL algorithms (e.g., PPO, DDPG), while retaining key advantages such as safety, feasibility, and reduced conservativeness.

The rest of the letter is organized as follows. In Section II, we provide preliminaries of CBF, RL, and HOCBF-based safe RL framework. We discuss two drawbacks of the framework in Section III. We propose the differential HOCBF-based safe RL framework in Section IV. Experiment results are shown and discussed in Section V. Finally, Section VI concludes the letter.

II. Preliminaries

In this section, we briefly introduce the CBF theory and underlying RL algorithm SAC. Then, we introduce HOCBF-based safe RL framework.

A. Markov Decision Process

Consider a Markov decision process (MDP) with control-affine system defined by a tuple $(\mathcal{X},\mathcal{U}, f, g, r, \gamma)$, where $\mathcal{X} \subset \mathbb{R}^n$ is the state space, $\mathcal{U} \subset \mathbb{R}^q$ is the control input space, $f : \mathbb{R}^n \to \mathbb{R}^n$, and $g : \mathbb{R}^n \to \mathbb{R}^{n\times q}$ are locally Lipschitz, $r : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$ is the reward function, and $\gamma \in (0, 1)$ is the discount factor. The control-affine system is given by

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t), \tag{1}$$

where $x(t) \in \mathcal{X}$ is the state, $u(t) \in \mathcal{U}$ is the control input. The objective of the MDP is to find an optimal control policy that maximizes the cumulative reward: $\sum_{t=0}^{\infty} \gamma^t r(x(t), u(t))$.

B. Control Barrier Function

Consider the safe set $\mathcal{C}$ defined as follows

$$\mathcal{C} = \{x(t) \in \mathcal{X} : h(x(t)) \geq 0\}, \tag{2}$$

where $h : \mathcal{X} \to \mathbb{R}$ is a continuously differentiable function.

Definition 1. (Control Barrier Function): For system (1) and safe set $\mathcal{C}$ defined in (2), h is a CBF if there exists a class $\mathcal{K}$ function $\alpha(\cdot)$ such that

$$L_f h(x(t)) + L_g h(x(t))u(t) + \alpha(h(x(t))) \geq 0, \tag{3}$$

where L_f, L_g denote the Lie derivatives along f and g, respectively.

For a function h with relative degree m $(m \geq 2)$, $u(t)$ does not explicitly show up in (3). To solve $u(t)$, starting with $\psi_0(x(t)) = h(x(t))$, we define a series of functions $\psi_i(x(t)), i \in \{1, 2, \ldots, m\}$

$$\psi_i\left(x(t)\right) = \dot{\psi}_{i-1}\left(x(t)\right) + \alpha_i(\psi_{i-1}\left(x(t)\right)), \tag{4}$$

where $\alpha_1(\cdot), \alpha_2(\cdot), \ldots, \alpha_m(\cdot)$ are class $\mathcal{K}$ functions. A sequence of set $\mathcal{C}_i$ associated with (4) is defined as

$$\mathcal{C}_i = \left\{x(t) \in \mathbb{R}^n \mid \psi_{i-1}\left(x(t)\right) \geq 0\right\}, i \in \{1, 2. \ldots, m\}. \tag{5}$$

Definition 2. (High-Order Control Barrier Function): A continuous function h is a HOCBF of relative degree m if there exists $\psi_i(x(t))$ defined by (4) and $\mathcal{C}_i$ defined by (5) such that

$$\begin{aligned}\psi_m\left(x(t)\right) &= L_f^m h(x(t)) + L_g L_f^{m-1} h(x(t))u(t) \\ &\quad + S(h(x(t))) + \alpha_m(\psi_{m-1}\left(x(t)\right)) \geq 0\end{aligned} \tag{6}$$

for all $x(t) \in \bigcap_{i=1}^m \mathcal{C}_i$. In (6), $L_f^m(L_g)$ denotes Lie derivatives along $f(g)$ m (one) times, and $S(h(x(t))) = \sum_{i=1}^{m-1} L_f^i(\alpha_{m-i} \circ \psi_{m-i-1})x(t)$, where $\circ$ denotes the composition of functions.

Lemma 1: [9] Given a HOCBF h and sets $\mathcal{C}_i$ defined as (5), if $x(0) \in \bigcap_{i=1}^m \mathcal{C}_i$, any $u(t)$ satisfying (6) renders the set $\bigcap_{i=1}^m \mathcal{C}_i$ forward invariant for system (1).

Remark 1: To simplify (6), class $\mathcal{K}$ functions $\alpha_i(\cdot), i \in \{1, 2. \ldots, m\}$ in (6) can be selected as linear functions with factor α_i.

C. Reinforcement Learning

Reinforcement learning aims to learn a policy by maximizing the long-term reward in an MDP. Let Δt be the time step interval, with t_k $(k \in \mathbb{N})$ as the sampling instants and $t_{k+1} = t_k + \Delta t$. The state and input at t_k are denoted by x_k and u_k, respectively. Discretization does not affect the CBF theory in Section II-B. Recent policy-gradient RL algorithms like SAC, PPO, and DDPG have achieved outstanding performance in continuous control. In this letter, we select SAC as the underlying RL algorithm for its state-of-the-art sample efficiency [17]. SAC relies on an actor-critic structure with parameterized Q-function $Q_\theta(x_k, u_k)$, target Q-function $Q_{\bar{\theta}}(x_k, u_k)$, and policy $\pi_\phi(u_k|x_k)$. The Q-function parameters are trained to minimize the following critic loss

$$J_Q(\theta) = \mathbb{E}_{(x_k, u_k^{RL}) \sim \mathcal{D}_R}\left[\frac{1}{2}\left(Q_\theta(x_k, u_k) - \left(r(x_k, u_k^{RL})\right.\right.\right.$$

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$$+\gamma\mathbb{E}_{u_{k+1}^{RL}\sim\pi_\phi}\left[Q_{\bar{\theta}}(x_{k+1},u_{k+1}^{RL})+\alpha_e\mathcal{H}(u_{k+1}^{RL}|x_{k+1})\right]\Big)\Big)^2\Bigg], \tag{7}$$

where $\mathcal{D}_R$ is the replay buffer, α_e is the temperature parameter, and the entropy term is defined as $\mathcal{H}(u_{k+1}^{RL}|x_{k+1}) = -\log\pi_\phi(u_{k+1}^{RL}|x_{k+1})$. The RL policy u_k^{RL} is parameterized using a neural network

$$u_k^{RL} = f_\phi(\epsilon_k; x_k) \tag{8}$$

where ϵ_t is the noise vector sampled from some distribution (e.g., spherical Gaussian). The policy parameters can be trained by minimizing the following policy loss

$$J_\pi(\phi) = \mathbb{E}_{x_k\sim\mathcal{D}_R,\epsilon_k\sim\mathcal{N}}\left[\alpha_e\log\pi_\phi(u_k^{RL}|x_k) - Q_\theta(x_k,u_k^{RL})\right]. \tag{9}$$

RL aims to maximize cumulative rewards but may lead to unsafe actions. To ensure safety, CBF-based safe RL methods have been studied in [10], [11], ensuring the RL agent satisfies safety constraints during training and deployment. Since HOCBF generalizes relative degree one CBF, CBF-based safe RL can be easily extended to HOCBF-based safe RL for safety constraints of relative degree m ($m \geq 2$).

D. HOCBF-Based Safe RL

In HOCBF-based safe RL, a CBF-based policy u_k^{CBF} is designed to minimally alter the RL policy u_k^{RL} to ensure safety. For system (1) and safe set $\mathcal{C}$ defined in (2), defining $u_k = u_k^{RL} + u_k^{CBF}$, the final safe action u_k is given by solving the following QP

$$\min_{u_k,\epsilon_s} \quad ||u_k - u_k^{RL}||_2 + K_\epsilon\epsilon_s^2 \tag{10a}$$

$$\text{s.t. } L_f^m h(x_k) + L_g L_f^{m-1}h(x_k)u_k + S(h(x_k)) + \alpha_m(\psi_{m-1}(x_k)) \geq \epsilon_s, \tag{10b}$$

$$u_{\min} \leq u_k \leq u_{\max}, \tag{10c}$$

where $\epsilon_s \in \mathbb{R}$ is a slack-variable utilized to ensure the feasibility of the QP, $K_\epsilon > 0$ is a large weighting term to avoid unnecessary safety violations, $u_{\min}$ and $u_{\max}$ denote the lower and upper bound of u_k. According to Lemma 1, the safety of HOCBF-based safe RL is ensured by the following Lemma.

Lemma 2: For system (1), given the HOCBF h and sets $\mathcal{C}_i$ defined as (5), if there exists a solution to (10) with $\epsilon_s = 0$, then the safe action u_k renders the set $\bigcap_{i=1}^m \mathcal{C}_i$ forward invariant.

III. Motivation

In HOCBF-based safe RL framework, the QP (10) is efficiently solved, ensuring safety during learning process. However, directly integrating the HOCBF constraint with RL presents challenges on conservativeness, and feasibility.

A. Conservativeness

As highlighted in [14], [15], the HOCBF constraint with manually designed $\alpha_i(\cdot)$ ensures safety but in a conservative way. The conservativeness extremely limits the performance of safe RL. Specifically, If $\alpha_i(\cdot)$ is too flat, the HOCBF constraint becomes over-conservative, activating prematurely when the state is far from the safety boundary. This reduces long-term rewards, as the agent may maintain unnecessarily large distances from obstacles, increasing travel distance. Conversely, if $\alpha_i(\cdot)$ is too steep, the HOCBF constraint becomes least-conservative, activating only near the safety boundary. This requires abrupt actions to ensure safety. For instance, the agent may require a sharp turn near obstacles to avoid collisions, but such abrupt signals are challenging for actuators to track accurately, increasing the risk of collisions and actuator damage. Therefore, we aim to design appropriate $\alpha_i(\cdot)$ to adjust the conservativeness of HOCBF constraint, thereby achieving the ideal performance of safe RL, i.e., obtaining large long-term rewards while avoiding abrupt changes in safe action.

The recently proposed adaptive CBFs [14] addressed the conservativeness by multiplying $\alpha_i(\cdot)$ in HOCBF with non-negative penalty functions $c_i(x_k)$, then functions $\psi_i(x_k), i \in \{1, 2, \ldots, m-1\}$ in (4) are redefined as

$$\psi_i(x_k) = \dot{\psi}_{i-1}(x_k) + c_i(x_k)\alpha_i(\psi_{i-1}(x_k)), \tag{11}$$

and the HOCBF constraint (6) is redefined as

$$\psi_m(x_k) = L_f^m h(x_k) + L_g L_f^{m-1}h(x_k)u_k + S(h(x_k)) + c_m(x_k)\alpha_m(\psi_{m-1}(x_k)) \geq 0. \tag{12}$$

By multiplying (relaxing) $\alpha_i(\cdot)$ with $c_i(x_k)$, [14] has proved that the satisfaction of (12) is a necessary and sufficient condition for the satisfaction of the original safety constraint $h(x_k) \geq 0$. In [14], $c_i(x_k)$ is derived from an auxiliary system, but the design of the system is difficult. To determine the variation of $c_i(x_k)$, the BarrierNet is proposed in [15], where penalty functions are represented by a neural network (NN), which can be trained on a dataset generated by the optimal controller (e.g.MPC). Although BarrierNet handles the conservativeness, it cannot be directly extended to HOCBF-based safe RL because the training relies on the nominal optimal controller, which is unavailable in safe RL. In this letter, we introduce NN-based penalty functions into HOCBF constraint and form a differential HOCBF-based QP. Subsequently, the NN-based penalty functions are trained aiming to achieving the ideal performance of safe RL. In the proposed approach, an optimal control dataset is not required; instead, the training only uses data generated by the RL exploration.

B. Feasibility

A slack variable ϵ_s is introduced to ensure the feasibility of (10). However, this results in a safety constraint violation if a nonzero ϵ_s is solved. To address the issue, we calculate a set for NN-based penalty functions, and prove that, under certain assumptions, there always exists NN-based penalty functions within the set such that the QP is feasible. Therefore, in the proposed framework, the slack variable is unnecessary, and the QP remains feasible even as the bounded input constraint is continuously tightened.

IV. Differential HOCBF-Based Safe RL

In this section, we introduce the differential HOCBF-based safe RL framework. Firstly, we employ a neural network to represent penalty functions and propose the differential HOCBF constraint. Next, we formulate a differential HOCBF-based QP as a trainable safety layer to guarantee safety. To ensure the feasibility of the QP, we calculate a set, and prove the QP is

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feasible if penalty functions remain within the set. Finally, we present the framework and outline the training process.

A. NN-Based Penalty Function

Define $P(x_k) = [p_1(x_k), p_2(x_k), \ldots, p_m(x_k)]^{\mathrm{T}}$, the penalty function $p_i(x_k) \in \mathbb{R}, i \in \{1, 2, \ldots, m\}$ is parameterized using the following Gaussian policy neural network

$$P(x_k) = f_\psi(\eta_k; x_k), \tag{13}$$

where η_k is the noise vector sampled from some distribution. The neural network in (13) is an MLP with n dimensional inputs, two hidden layers of 256 neurons followed by the ReLu activation function, and two separate output layers respectively producing the mean and standard deviation of an m-dimensional Gaussian distribution. The penalty function $P(x_k)$ is sampled from this distribution using the reparameterization trick. As discussed in Section III-A, $p_i(x_k) \to 0$ corresponds to an extremely over-conservative constraint, which yields very low long-term rewards. Conversely, $p_i(x_k) \to +\infty$ represents a highly least-conservative constraint, leading to abrupt control actions for ensuring safety. To avoid the above extreme cases during training, $p_i(x_k)$ is limited within a set

$$p_i(x_k) \in \mathcal{P} = \left\{p \mid 0 < \underline{p} \le p \le \overline{p}\right\}, i \in \{1, 2, \ldots, m\} \tag{14}$$

where $\underline{p}$ is the lower bound representing a relatively over-conservative constraint and $\overline{p}$ is the upper bound corresponding to a relatively least-conservative constraint. Similar to (11), utilizing NN-based penalty function, we define a series of functions $\boldsymbol{\psi}_i(x_k), i \in \{1, 2, \ldots, m\}$

$$\boldsymbol{\psi}_i(x_k) = \dot{\boldsymbol{\psi}}_{i-1}(x_k) + p_i(x_k)\alpha_i(\boldsymbol{\psi}_{i-1}(x_k)), \tag{15}$$

with $\boldsymbol{\psi}_0(x_k) = h(x_k)$. A sequence of set $\boldsymbol{C}_i$ associated with (15) is defined as

$$\boldsymbol{C}_i = \{x_k \in \mathbb{R}^n \mid \boldsymbol{\psi}_{i-1}(x_k) \ge 0\}, i \in \{1, 2 \ldots, m\}. \tag{16}$$

The differential HOCBF constraint is defined as follows

$$\begin{aligned} &L_f^m h(x_k) + L_g L_f^{m-1} h(x_k) u_k + S(h(x_k)) \\ &\quad + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)) \ge 0. \end{aligned} \tag{17}$$

As discussed in Section III-A, the penalty functions $p_i(x_k)$ can multiply (relax) the manually designed $\alpha_i(\cdot)$, thereby allowing dynamic adjustment of the conservativeness. Based on (17), differential HOCBF-based QP is proposed in the next section.

B. Differential HOCBF-Based QP

Given RL policy u_k^{RL} and current safe state x_k, using differential HOCBF constraint (17), we formulate the differential HOCBF-based QP as follows

$$\min_{u_k} \quad ||u_k - u_k^{RL}||_2 \tag{18a}$$

$$\text{s.t. } Gu_k \le H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)), \tag{18b}$$

$$u_{\min} \le u_k \le u_{\max}, \tag{18c}$$

where $G = -L_g L_f^{m-1} h(x_k), H = L_f^m h(x_k) + S(h(x_k))$. As illustrated in Fig. 1, the QP in (18) is implemented as a differentiable layer [18], whose output is the QP solution. According to [18], this layer can efficiently handle large-scale QPs and is differentiable with respect to the parameters ψ and ϕ.

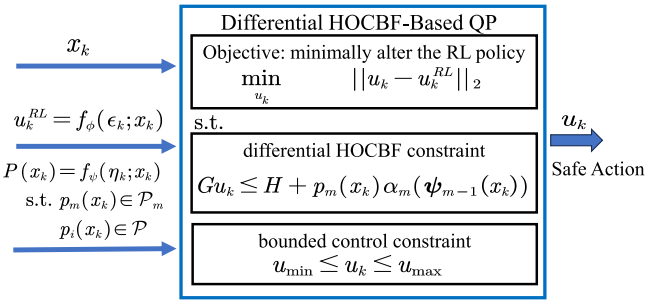

Fig. 1. The structure of differential HOCBF-based QP.

In (18), the QP may be infeasible when both the differential HOCBF constraint (18b) and the bounded input constraint (18c) are considered. To address the infeasibility, we calculate a set $\mathcal{F}$, and the following theorem proves that if $p_m(x_k) \in \mathcal{F}$, the QP (18) is feasible.

Theorem 1: Assume $u_{\min} < u_k^{RL} < u_{\max}$ and $\boldsymbol{\psi}_{m-1}(x_k) > 0$. If $p_m(x_k) \in \mathcal{F} = \{p \mid p_{\min} \le pG^{\mathrm{T}} \le p_{\max}\}$, where $p_{\min} = (GG^{\mathrm{T}}(u_{\min} - u_k^{RL}) + Gu_k^{RL}G^{\mathrm{T}} - HG^{\mathrm{T}})/\alpha_m(\boldsymbol{\psi}_{m-1}(x_k))$ and $p_{\max} = (GG^{\mathrm{T}}(u_{\max} - u_k^{RL}) + Gu_k^{RL}G^{\mathrm{T}} - HG^{\mathrm{T}})/\alpha_m(\boldsymbol{\psi}_{m-1}(x_k))$, then the QP (18) is feasible.

Proof: If the constraint (18b) is inactive, it is evident that the QP remains feasible. If the constraint (18b) becomes active, the QP might be infeasible. In this case, we define a new QP as follows:

$$\min_{u_k} \quad ||u_k - u_k^{RL}||_2 \tag{19a}$$

$$\text{s.t. } Gu_k \le H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)), \tag{19b}$$

Let u_k^* denote the optimal solution of QP (19). A sufficient condition for the feasibility of QP (18) is that u_k^* satisfies the bounded input constraint (18c). We define the Lagrangian function $\mathcal{L}$ for QP (19) as follows

$$\mathcal{L} = ||u_k - u_k^{RL}||_2 + \lambda(Gu_k - (H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)))). \tag{20}$$

According to the Karush–Kuhn–Tucker (KKT) conditions, when (19b) is active, u_k^* is the optimal solution of (19) if the following conditions are satisfied

$$\begin{aligned} &\frac{\partial \mathcal{L}}{\partial u_k^*} = 2(u_k^* - u_k^{RL}) + \lambda G^{\mathrm{T}} = 0, \\ &Gu_k^* - (H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k))) = 0. \end{aligned} \tag{21}$$

The solution to (21) is

$$\begin{aligned} &\lambda = 2\frac{Gu_k^{RL} - (H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)))}{GG^{\mathrm{T}}}, \\ &u_k^* = u_k^{RL} + \frac{H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)) - Gu_k^{RL}}{GG^{\mathrm{T}}}G^{\mathrm{T}}. \end{aligned} \tag{22}$$

Therefore, when (18b) is active, the QP (18) is feasible if the following condition is satisfied

$$u_{\min} \le u_k^* \le u_{\max}. \tag{23}$$

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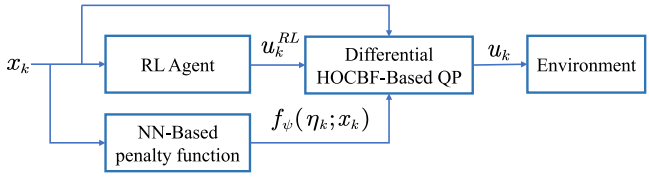

Fig. 2. The diagram of the proposed framework.

We find that if $u_{\min} < u_k^{RL} < u_{\max}$, there always exists $H + p_m(x_k)\alpha_m(\boldsymbol{\psi}_{m-1}(x_k)) - Gu_k^{RL}/(GG^{\mathrm{T}})$, such that (23) is satisfied. Therefore, if $p_m(x_k) \in \mathcal{F}$, then (23) is satisfied, i.e., the QP (18) is feasible.

Since $p_i(x_k)$ is also required to satisfy (14), we have

$$p_m(x_k) \in \mathcal{P}_m = \begin{cases} \mathcal{F} \cap \mathcal{P}, & \text{if } \mathcal{F} \cap \mathcal{P} \neq \emptyset, \\ \{p \mid p \in \mathcal{F}, p \neq 0\}, & \text{if } \mathcal{F} \cap \mathcal{P} = \emptyset. \end{cases}$$
$$p_i(x_k) \in \mathcal{P}, \qquad i \in \{1, 2, \ldots, m-1\}. \tag{24}$$

According to Theorem 1, $p_m(x_k) \in \mathcal{P}_m$ guarantees the feasibility of QP (18). Using tanh function in the final layer of $f_\psi(\eta_k; x_k)$, the penalty function $p_i(x_k), i \in \{1, 2, \ldots, m\}$ can be limited within corresponding set.

Theorem 2: Given a HOCBF h and sets $\boldsymbol{C}_i$ defined as (16), if the initial state remains within $\bigcap_{i=1}^m \boldsymbol{C}_i$ and (24) is satisfied, then the QP (18) is solved with unique solution u_k, and u_k renders the state within the safe set $\mathcal{C}$.

Proof: In (18), the cost function is a positive-definite quadratic form and all constraints are linear. If (24) is satisfied, then the optimization problem (18) is a feasible convex QP, which is solved with a unique solution u_k. Next, we prove the safety by analyzing two cases. If $p_m(x_k) > 0$, $p_i(x_k)\alpha_i(\cdot)$ in $\boldsymbol{\psi}_i(x_k), i \in \{1, 2 \ldots, m\}$ can be considered as a new class $\mathcal{K}$ function. According to Lemma 1, u_k renders the state within the safe set $\mathcal{C}$. If $p_m(x_k) < 0$, the satisfaction of (18b) implies $Gu_k \leq H + \alpha_m(\boldsymbol{\psi}_{m-1}(x_k))$, and $p_i(x_k)\alpha_i(\cdot)$ in $\boldsymbol{\psi}_i(x_k), i \in \{1, 2 \ldots, m-1\}$ can also be considered as a new class $\mathcal{K}$ function. According to Lemma 1, u_k renders the state within the safe set $\mathcal{C}$.

Based on the above analysis, we have the following conclusions:

1) Penalty function $p_i(x_k)$ can adjust the conservativeness of differential HOCBF constraint (17), thereby achieving ideal control performance.
2) If the constraint (24) is satisfied, the QP (18) is feasible, and its solution u_k renders the state within the safe set.

In the next section, we aim to learn appropriate $p_i(x_k)$ to achieve the ideal performance. To address this, we propose the differential HOCBF-based safe RL framework.

C. Differential HOCBF-Based Safe RL

The structure of the proposed framework is illustrated in Fig. 2. Based on the current state x_k, u_k^{RL} and $f_\psi(\eta_k; x_k)$ are provided by RL agent and NN-based penalty functions respectively, then a differential HOCBF-based QP is formulated and the safe action u_k is solved to interact with the environment.

To solve the safe control action that maximizes long-term rewards while preventing abrupt action changes, we simultaneously train both the RL policy and the NN-based penalty functions. Specifically, the RL policy is trained following the SAC algorithm as outlined in Section II-C. The Gaussian policy $f_\psi(\eta_k; x_k)$ is trained to generate appropriate state-dependent penalty function $p_i(x_k)$ to adaptively adjust the conservativeness of differential HOCBF constraint. This adjustment enables the differential HOCBF-based QP output optimal u_k for achieving ideal performance. The NN-based penalty functions are trained by minimizing the following penalty function loss

$$J_p(\psi) = \mathbb{E}_{x_k \sim \mathcal{D}_R, \eta_k \sim \mathcal{N}} \left[-\alpha_e \mathcal{H}(f_\psi(\eta_k; x_k)) - Q_\theta(x_k, u_k) + B(u_k, u_{k-1}) \right]. \tag{25}$$

The penalty function loss (25) includes three components. The entropy term $-\alpha_e \mathcal{H}(f_\psi(\eta_k; x_k))$ is used to encourage exploration by promoting stochasticity in $f_\psi(\eta_k; x_k)$. Note that $f_\psi(\eta_k; x_k)$ is required to satisfy (24) during the stochastic exploration to ensure the feasibility of QP (18). The Q-function term $Q_\theta(x_k, u_k)$ estimates the expected cumulative reward of taking the safe action u_k at state x_k. As u_k is solved by QP (18) formulated by $f_\psi(\eta_k; x_k)$, we aim to select an appropriate $f_\psi(\eta_k; x_k)$, so that the agent can obtain the maximum $Q_\theta(x_k, u_k)$ by taking safe action u_k. The abrupt changes penalty term $B(u_k, u_{k-1})$ is a custom-defined function to penalize the abrupt changes in safe action. Therefore, minimizing the loss function (25) aims to explore an appropriate $f_\psi(\eta_k; x_k)$ that maximizes long-term rewards and avoids abrupt changes in safe action while ensuring the feasibility of the QP (18). Compared with HOCBF-based safe RL using manually designed $\alpha_i(\cdot)$, the proposed framework achieves ideal performance by training $f_\psi(\eta_k; x_k)$. We summarize the training process in Algorithm 1.

Algorithm 1: Differential HOCBF-Based Safe RL.

1: Initialization
Initialize Q-function Q_θ, RL policy π_ϕ, NN-based penalty function f_ψ, experience replay buffer $\mathcal{D}_R$, maximal RL training episodes K, learning rate $\lambda_\theta, \lambda_\psi, \lambda_\phi$
2:Differential HOCBF-based Safe RL Training
for episode $= 0, 1, \ldots, K$ **do**
  Reset the environment
  while not done **do**
    Compute $\mathcal{P}_m, \mathcal{P}$ based on current state x_k
    Sample RL action u_k^{RL}, penalty function $f_\psi(\eta_k; x_k)$
    Solve safe action u_k from QP (18)
    Deploy safe action u_k
    Observe x_{k+1}, r_k
    Store transition (x_k, u_k, x_{k+1}, r_k) in $\mathcal{D}_\mathcal{R}$
    Sample a random mini-batch of transitions from $\mathcal{D}_\mathcal{R}$
    $\theta \leftarrow \theta - \lambda_\theta \nabla_\theta J_Q(\theta)$
    $\phi \leftarrow \phi - \lambda_\phi \nabla_\phi J_\pi(\phi)$
    $\psi \leftarrow \psi - \lambda_\psi \nabla_\psi J_p(\psi)$
  end while
end for

Remark 2: The proposed differential HOCBF-based QP is an expandable framework that can be readily integrated with other policy gradient-based RL algorithms (e.g., PPO, DDPG). Consider the integration with PPO as an example. The Algorithm 1 can be modified by (i) replacing SAC with PPO, and (ii) $Q_\theta(x_k, u_k)$ in (25) is provided by PPO.

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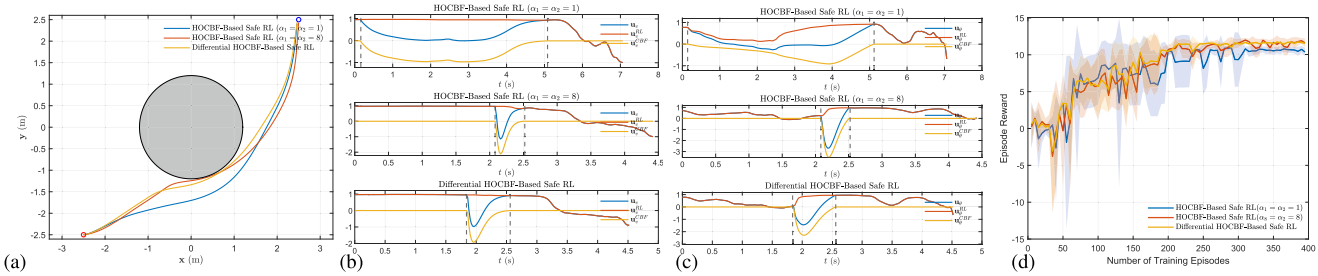

Fig. 3. Control performance of differential HOCBF-based safe RL and HOCBF-based safe RL under two distinct parameter settings in Scenario 1. The black dotted line represents the time interval when the HOCBF constraint or differential HOCBF constraint is active. (a) Robot trajectory. (b) Control input u_v. (c) Control input u_θ. (d) Episode Reward.

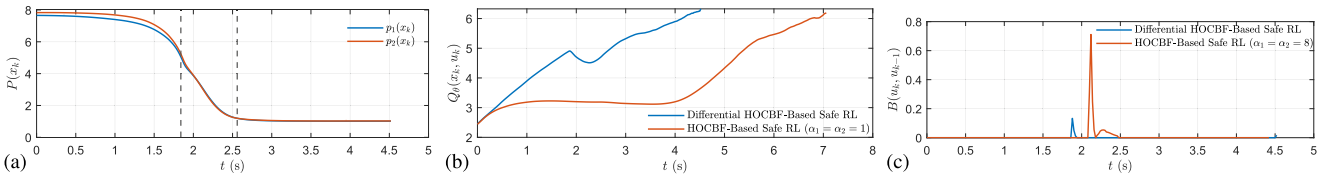

Fig. 4. (a) The profiles of penalty functions $p_1(x_k)$ and $p_2(x_k)$ learned by differential HOCBF-based safe RL in Scenario 1. (b) Comparison of Q-function term $Q_\theta(x_k, u_k)$ for differential HOCBF-based safe RL and HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 1$) in Scenario 1. (c) Comparison of abrupt changes penalty term $B(u_k, u_{k-1})$ for differential HOCBF-based safe RL and HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 8$) in Scenario 1.

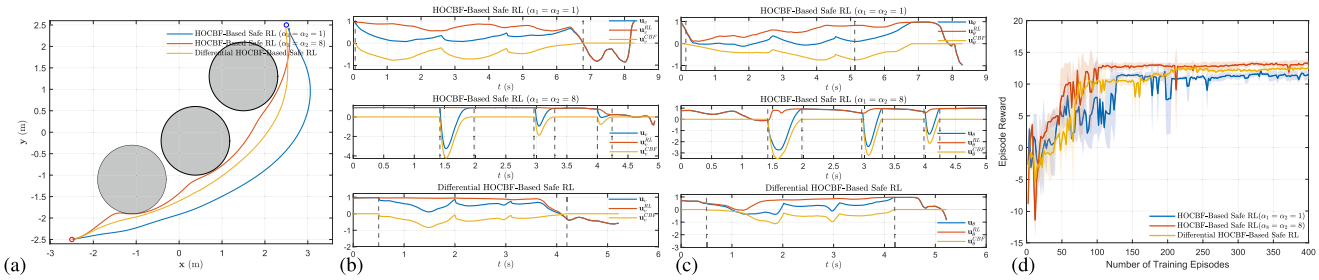

Fig. 5. Control performance of differential HOCBF-based safe RL and HOCBF-based safe RL under two distinct parameter settings in Scenario 2. (a) Robot trajectory. (b) Control input u_v. (c) Control input u_θ. (d) Episode Reward.

V. Experiments

In this section, we evaluate the proposed framework using a wheeled mobile robot tasked with reaching a target location while avoiding obstacles. The first set of experiments demonstrates that the proposed approach effectively adjusts the conservativeness of the HOCBF constraint, thereby achieving ideal safe RL performance. The second set of experiments confirms that the QP in the proposed framework remains feasible even under progressively tighter input constraints.

A. Environment Setup

The wheeled mobile robot is modeled by the dynamic as

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{v}\cos(\boldsymbol{\theta}),\\ \dot{\mathbf{y}} &= \mathbf{v}\sin(\boldsymbol{\theta}),\\ \dot{\mathbf{v}} &= \mathbf{u}_v,\\ \dot{\boldsymbol{\theta}} &= \mathbf{u}_\theta,\end{aligned} \tag{26}$$

where $x_k := [\mathbf{x}, \mathbf{y}, \mathbf{v}, \boldsymbol{\theta}]^{\mathrm{T}}, u_k := [\mathbf{u}_v, \mathbf{u}_\theta]^{\mathrm{T}}$, $\mathbf{x}, \mathbf{y}$ represent the 2-D coordinates, $\mathbf{v}$ denote the forward velocity, and $\boldsymbol{\theta}$ is the yaw angle of the robot. The control input $\mathbf{u}_v, \mathbf{u}_\theta$ represent acceleration and yaw angle velocity respectively. The reward r_k is designed as follows: $r_k = d_{k-1} - d_k$ if the goal is not reached, and $r_k = d_{k-1} - d_k + 1 + \frac{1.2T}{k_r}$ if the goal is reached, where d_k represents the distance between the robot and the goal location at sampling instant t_k, T is the maximum time steps per episode, k_r is the time step when the robot reaches the goal. The bounded inputs constraint is defined as $\mathbf{u}_v^{min} \leq \mathbf{u}_v \leq \mathbf{u}_v^{max}$, and $\mathbf{u}_\theta^{min} \leq \mathbf{u}_\theta \leq \mathbf{u}_\theta^{max}$.

B. Differential HOCBF-Based Safe RL Design

The robot is required to avoid the circular obstacle in its path. The safety constraint is defined as

$$h(x_k) = (\mathbf{x} - \mathbf{x}_{ob})^2 + (\mathbf{y} - \mathbf{y}_{ob})^2 - R_s^2 \geq 0, \tag{27}$$

where $(\mathbf{x}_{ob}, \mathbf{y}_{ob})$ is the position of the obstacle, R_s is the safety distance. The relative degree of (27) is 2 with respect to the dynamic (26). According to Definition 2, the HOCBF constraint

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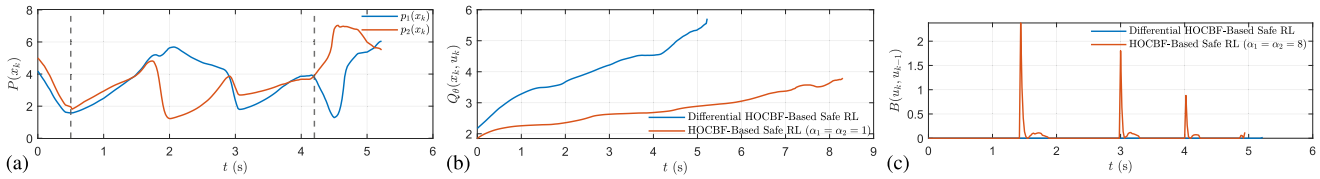

Fig. 6. (a) The profiles of penalty functions $p_1(x_k)$ and $p_2(x_k)$ learned by differential HOCBF-based safe RL in Scenario 2. (b) Comparison of Q-function term $Q_\theta(x_k, u_k)$ for differential HOCBF-based safe RL and HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 1$) in Scenario 2. (c) Comparison of abrupt changes penalty term $B(u_k, u_{k-1})$ for differential HOCBF-based safe RL and HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 8$) in Scenario 2.

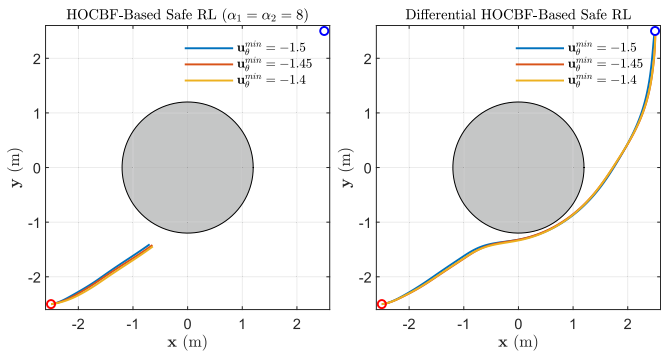

Fig. 7. The QP in HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 8$) becomes infeasible, making the trajectory unsolvable. In contrast, the QP in differential HOCBF-based safe RL remains feasible under progressively tightened input constraints.
.

(choose $\alpha_1(\cdot), \alpha_2(\cdot)$ as linear functions with factors α_1, α_2) is defined as

$$-L_g L_f h(x_k) u_k \leq L_f^2 h(x_k) + (\alpha_1 + \alpha_2) L_f h(x_k) + \alpha_1 \alpha_2 h(x_k), \tag{28}$$

where $L_g L_f h(x_k) = [-2(\mathbf{x} - \mathbf{x}_{ob})\mathbf{v} \sin \boldsymbol{\theta} + 2(\mathbf{y} - \mathbf{y}_{ob})\mathbf{v} \cos \boldsymbol{\theta}, 2(\mathbf{x} - \mathbf{x}_{ob}) \cos \boldsymbol{\theta} + 2(\mathbf{y} - \mathbf{y}_{ob}) \sin \boldsymbol{\theta}]$, $L_f^2 h(x_k) = 2\mathbf{v}^2$, $L_f h(x_k) = 2(\mathbf{x} - \mathbf{x}_{ob})\mathbf{v} \cos \boldsymbol{\theta} + 2(\mathbf{y} - \mathbf{y}_{ob})\mathbf{v} \sin \boldsymbol{\theta}$. As a contrast, in the proposed framework, we introduce NN-based penalty functions $p_1(x_k), p_2(x_k)$, and define the differential HOCBF constraint (choose $\alpha_1(\cdot), \alpha_2(\cdot)$ as linear functions with factors $\alpha_1 = \alpha_2 = 1$) as

$$-L_g L_f h(x_k) u_k \leq L_f^2 h(x_k) + (p_1(x_k) + p_2(x_k)) L_f h(x_k) + p_1(x_k) p_2(x_k) h(x_k). \tag{29}$$

We set $\underline{p} = 1, \overline{p} = 8$. The goal is to train $p_1(x_k), p_2(x_k)$, and RL policy simultaneously to achieve an ideal performance, i.e., maximizing long-term rewards while avoiding abrupt changes in safe action. The abrupt changes penalty term $B(u_k, u_{k-1})$ in (25) is defined as $B(u_k, u_{k-1}) = 0$ if $||u_k - u_{k-1}||_2 < 0.35$, and $B(u_k, u_{k-1}) = 0.2||u_k - u_{k-1}||_2$ if $||u_k - u_{k-1}||_2 \geq 0.35$.

C. Results and Discussion

1) Conservativeness: We compare differential HOCBF-based safe RL and HOCBF-based safe RL under two parameter settings: over-conservative HOCBF constraint ($\alpha_1 = \alpha_2 = 1$) and least-conservative HOCBF constraint ($\alpha_1 = \alpha_2 = 8$). We apply the three methods to two scenarios: Scenario 1 - robot navigating around one obstacle, and Scenario 2 - robot navigating around three obstacles. Note that in both the training and testing phases, obstacle positions are randomly sampled from predefined regions according to a specific distribution. The results shown below are only one instance in the testing phase. For each scenario, ten independent training runs are conducted, and the mean and variance of the cumulative rewards are reported.

The results of Scenario 1 are shown in Figs. 3 and 4. For HOCBF-based safe RL with $\alpha_1 = \alpha_2 = 1$, the over-conservative HOCBF constraint forces the robot to maintain excessive distance from the obstacle (Fig. 3(a)), resulting in a longer trajectory and consequently lower rewards (Fig. 3(d)). In contrast, for HOCBF-based safe RL with $\alpha_1 = \alpha_2 = 8$, although achieving higher rewards (Fig. 3(d)), the least-conservative HOCBF constraint only activates in close proximity to the obstacle, causing abrupt changes in both $\mathbf{u}_v$ and $\mathbf{u}_\theta$ (Fig. 3(b)–(c)). Our proposed differential HOCBF-based safe RL framework addresses these drawbacks through learning $p_1(x_k)$ and $p_2(x_k)$ (Fig. 4(a)). Penalty functions $p_1(x_k)$ and $p_2(x_k)$ dynamically adjust the HOCBF constraint conservativeness as the robot approaches obstacles, providing two key advantages: (1) Compared to the over-conservative case ($\alpha_1 = \alpha_2 = 1$), our method solves for a suitable safe action u_k that aims to maximize $Q_\theta(x_k, u_k)$ (Fig. 4(b)), thereby achieving higher rewards approaching to the least-conservative case (Fig. 3(d)); (2) Compared to the least-conservative case ($\alpha_1 = \alpha_2 = 8$), $p_1(x_k)$ and $p_2(x_k)$ are trained to minimize the abrupt changes penalty term $B(u_k, u_{k-1})$ (Fig. 4(c)), effectively eliminating the abrupt changes in control, as observed in Fig. 3(b)-(c).

The results of Scenario 2 are presented in Figs. 5 and 6. For HOCBF-based safe RL with $\alpha_1 = \alpha_2 = 1$, simultaneously considering the three obstacles, the robot is controlled to maintain more excessive distance from obstacles (Fig. 5(a)), resulting in lower rewards (Fig. 5(d)). For HOCBF-based safe RL with $\alpha_1 = \alpha_2 = 8$, the robot closely follows obstacle boundaries during collision avoidance, resulting in significant abrupt changes in both $\mathbf{u}_v$ and $\mathbf{u}_\theta$ whenever an obstacle is encountered (Fig. 5(b)–(c)). The proposed differential HOCBF-based safe RL framework addresses these issues by training $p_1(x_k)$ and $p_2(x_k)$ (Fig. 6(a)) to dynamically adjust the HOCBF constraint conservativeness during obstacles avoidance. Compared to the over-conservative case ($\alpha_1 = \alpha_2 = 1$), this approach achieves significantly higher rewards (Fig. 5(d)) by maximizing the Q-function term $Q_\theta(x_k, u_k)$ (Fig. 4(b)). Compared to the least-conservative case ($\alpha_1 = \alpha_2 = 8$), although the reward is lower, the trajectory solved by the proposed framework is smoother, and abrupt control inputs changes are avoided. This is because the $p_1(x_k)$ and $p_2(x_k)$ are trained to minimize the abrupt changes penalty term $B(u_k, u_{k-1})$ when the robot approaches each obstacle (Fig. 6(c)).

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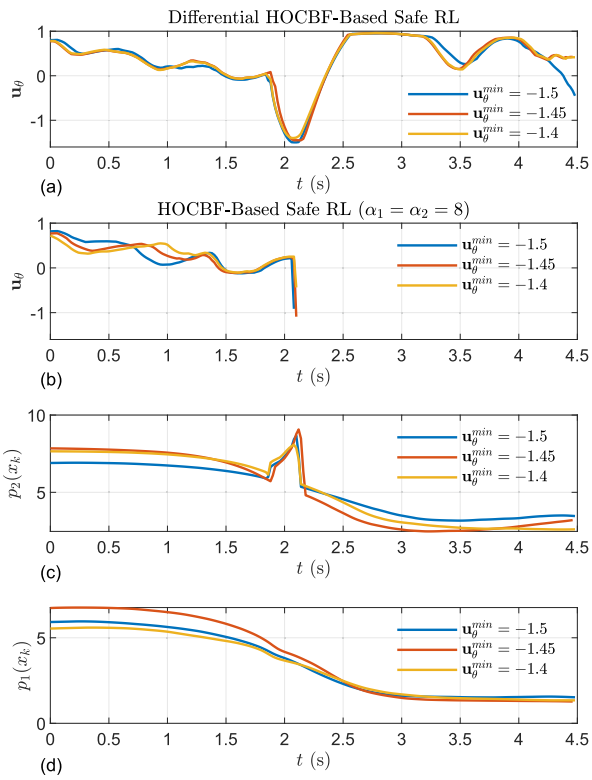

Fig. 8. As $\mathbf{u}_\theta$ approaches its lower bound $\mathbf{u}_\theta^{\min}$, $p_2(x_k)$ learns to dynamically adapt to satisfy the constraint $p_2(x_k) \in \mathcal{P}_2$, thereby ensuring the feasibility. (a) The QP in the differential HOCBF-based safe RL remains feasible when the bounded input constraint is progressively tightened. (b) The QP in HOCBF-based safe RL ($\alpha_1 = \alpha_2 = 8$) encounters infeasibility issue when the input bound is progressively tightened. (c) The profile of penalty function $p_2(x_k)$. (d) The profile of penalty function $p_1(x_k)$. .

2) Feasibility: We utilize Scenario 1 to evaluate the feasibility of the QP in both differential HOCBF-based safe RL and HOCBF-based safe RL with $\alpha_1 = \alpha_2 = 8$ under progressively tightened input bounds. As shown in Fig. 8(b), the QP in the HOCBF-based safe RL becomes infeasible when the slack variable ϵ_s is set to zero to ensure strict safety. The infeasible issues leads the safe obstacle avoidance trajectory is unsolvable (Fig. 7). In contrast, Fig. 8(a) illustrates that the QP in the proposed framework remains feasible even as the input bounds are progressively tightened. As demonstrated in Fig. 8(c), precisely when $\mathbf{u}_\theta$ approaches its lower bound, the learned $p_2(x_k)$ adaptively adjusts to ensure $p_2(x_k) \in \mathcal{P}_2$ is satisfied, thereby maintaining the feasibility.

VI. Conclusion

We propose the differential HOCBF-based safe RL framework to address the conservativeness and feasibility issues. We introduce the NN-based penalty functions into the HOCBF constraint, and train the penalty functions. The experiment results verify the effectiveness of the proposed framework.

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